

**1. Explain Bayesian Statistics with reference to Bayes Theorem**

**The Basics of Bayesian Statistics**

- ❖ Bayesian statistics is a branch of statistics that deals with uncertainty by incorporating prior knowledge or beliefs into the analysis of data.
- ❖ It is named after Thomas Bayes, an 18th-century mathematician and theologian. The fundamental concept in Bayesian statistics is the Bayes' theorem, which describes how to update our beliefs in light of new evidence. Here are the basics of Bayesian statistics:
- ❖ Bayesian statistics is a powerful approach for making statistical inferences, especially when dealing with small or complex data sets and when incorporating prior information is valuable or necessary.
- ❖ However, the choice of prior can be subjective and influence results, so it's essential to carefully consider and justify your prior assumptions in Bayesian analysis.
- ❖ Bayesian statistics is a statistical theory and approach to data analysis that uses Bayes' theorem to describe the probability of an event based on previous knowledge and observed and unobserved parameters.

**Bayes' Theorem**

- ❖ Bayes' Theorem, named after 18th-century British mathematician Thomas Bayes, is a mathematical formula for determining conditional probability. Conditional probability is the likelihood of an outcome occurring based on a previous outcome in similar circumstances.

- ❖ The Bayes theorem is frequently referred to as the Bayes rule or Bayes Law. One of the most well-known theories in machine learning, the Bayes theorem helps determine the likelihood that one event will occur with unclear information while another has already happened.
- ❖ Bayes' theorem describes the probability of occurrence of an event related to any condition. It is also considered for the case of conditional probability. Bayes theorem is also known as the formula for the probability of “causes”

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

#### **Bayes' Theorem:**

- ❖ At its core, Bayesian statistics is built upon Bayes' theorem, which relates conditional probabilities. The theorem can be expressed as follows:

$$❖ P(A | B) = [P(B | A) * P(A)] / P(B)$$

- ❖  $P(A | B)$  represents the probability of event A occurring given that event B has occurred.
- ❖  $P(B | A)$  is the probability of event B occurring given that event A has occurred.
- ❖  $P(A)$  and  $P(B)$  are the marginal probabilities of events A and B, respectively.

#### **In the context of Bayesian statistics:**

- ❖  $P(A | B)$  is the posterior probability, which represents our updated belief in A after observing B.
- ❖  $P(B | A)$  is the likelihood, describing the probability of observing B if A is true.
- ❖  $P(A)$  is the prior probability, representing our initial belief in A before considering B.
- ❖  $P(B)$  is the marginal likelihood or evidence, the probability of observing B without considering A.

## 2. Explain the steps involved in build an ML Algorithm.

### ❖ Building a Machine Learning Algorithm

- ❖ Nearly all deep learning algorithms can be described as particular instances of a fairly simple recipe: combine a specification of a dataset, a cost function, an optimization procedure and a model.
- ❖ For example, the linear regression algorithm combines a dataset consisting of  $\mathbf{X}$  and  $y$ , the cost function.

$$J(\mathbf{w}, b) = -\mathbb{E}_{\mathbf{x}, y \sim \hat{p}_{\text{data}}} \log p_{\text{model}}(y | \mathbf{x}),$$

- ❖ The model specification  $p_{\text{model}}(y | \mathbf{x}) = \mathcal{N}(y; \mathbf{x}^T \mathbf{w} + b, 1)$ , and, in most cases, the optimization algorithm defined by solving for where the gradient of the cost (slope) is zero using the normal equations.
- ❖ By realizing that we can replace any of these components mostly independently from the others, we can obtain a very wide variety of algorithms.
- ❖ The cost function typically includes at least one term that causes the learning process to perform statistical estimation. The most common cost function is the negative log-likelihood, so that minimizing the cost function causes maximum likelihood estimation
- ❖ The cost function may also include additional terms, such as regularization terms. For example, we can add weight decay to the linear regression cost function to obtain.

$$J(\mathbf{w}, b) = \lambda \|\mathbf{w}\|_2^2 - \mathbb{E}_{\mathbf{x}, y \sim p_{\text{data}}} \log p_{\text{model}}(y | \mathbf{x}).$$

- ❖ This still allows closed-form optimization.
- ❖ If we change the model to be nonlinear, then most cost functions can no longer be optimized in closed form. This requires us to choose an iterative numerical optimization procedure, such as gradient descent.
- ❖ The recipe for constructing a learning algorithm by combining models, costs, and optimization algorithms supports both supervised and unsupervised learning.
- ❖ The linear regression example shows how to support supervised learning.
- ❖ Unsupervised learning can be supported by defining a dataset that contains only  $\mathbf{X}$  and providing an appropriate unsupervised cost and model.
- ❖ For example, we can obtain the first PCA vector by specifying that our loss function is

$$J(\mathbf{w}) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} \|\mathbf{x} - r(\mathbf{x}; \mathbf{w})\|_2^2$$

- ❖ While our model is defined to have  $\mathbf{w}$  with norm one and reconstruction function

$$r(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} \mathbf{w}.$$

- ❖ In some cases, the cost function may be a function that we cannot actually evaluate, for computational reasons.
- ❖ In these cases, we can still approximately minimize it using iterative numerical optimization so long as we have some way of approximating its gradients.
- ❖ Most machine learning algorithms make use of this recipe, though it may not immediately be obvious.
- ❖ If a machine learning algorithm seems especially unique or hand-designed, it can usually be understood as using a special-case optimizer.

- ❖ Some models such as decision trees or k-means require special-case optimizers because their cost functions have flat regions that make them inappropriate for minimization by gradient-based optimizers.
- ❖ Recognizing that most machine learning algorithms can be described using this recipe helps to see the different algorithms as part of a taxonomy of methods for doing related tasks that work for similar reasons, rather than as a long list of algorithms that each have separate justifications

### 3. Explain the key differences between Machine Learning and Deep Learning with examples.

| Aspect                            | Machine Learning                                               | Deep Learning                                                         |
|-----------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Definition</b>                 | Subset of AI that learns patterns from data                    | Subset of ML using deep neural networks                               |
| <b>Data Dependency</b>            | Works well with smaller, structured datasets                   | Requires large amounts of data, works with unstructured data          |
| <b>Feature Engineering</b>        | Requires manual feature extraction and selection               | Automatically extracts features from raw data                         |
| <b>Model Complexity</b>           | Simpler, more interpretable models                             | Complex models with multiple layers, harder to interpret              |
| <b>Computational Requirements</b> | Less computational power needed                                | High computational power, often relies on GPUs                        |
| <b>Applications</b>               | Recommendation systems, fraud detection, customer segmentation | Image/video analysis, speech recognition, natural language processing |

| <b>Aspect</b>                 | <b>Machine Learning</b>                       | <b>Deep Learning</b>                                                   |
|-------------------------------|-----------------------------------------------|------------------------------------------------------------------------|
| <b>Training Time</b>          | Shorter training times                        | Longer training times due to complexity and data volume                |
| <b>Example Algorithms</b>     | Linear regression, decision trees, SVM, k-NN  | Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs) |
| <b>Performance with Scale</b> | May not improve significantly with more data  | Performance improves with more data and deeper models                  |
| <b>Interpretability</b>       | Generally more interpretable and explainable. | Often considered a "black box" with less interpretability.             |

### Difference between Overfitting, Underfitting and Capacity in detail.

| S.No | Aspect                   | Overfitting                                                                                                    | Underfitting                                                                                | Model Capacity                                                                          |
|------|--------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| 1    | <b>Definition</b>        | Model learns the training data <b>too well</b> , including noise and outliers, leading to poor generalization. | Model <b>fails to learn</b> the underlying structure in the training data.                  | Model's <b>flexibility to learn</b> different types of functions.                       |
| 2    | <b>Training Accuracy</b> | Very high. The model performs extremely well on the training dataset.                                          | Low. The model performs poorly even on training data.                                       | Can be adjusted to be low, moderate, or high.                                           |
| 3    | <b>Test Accuracy</b>     | Low. The model performs poorly on unseen or test data due to memorization of noise.                            | Low. The model also performs poorly on test data due to inability to capture relationships. | Proper capacity leads to good performance on both training and test data.               |
| 4    | <b>Bias</b>              | <b>Low bias</b> – Model is highly flexible and assumes fewer restrictions on the data.                         | <b>High bias</b> – Model is too rigid and assumes too many simplifications.                 | Capacity affects bias — low capacity leads to high bias, and high capacity to low bias. |
| 5    | <b>Variance</b>          | <b>High variance</b> – Small changes in input lead to large changes in output.                                 | <b>Low variance</b> – Predictions do not change much with changes in input.                 | High-capacity models have high variance; low-capacity models have low variance.         |

| S.No | Aspect                       | Overfitting                                                                                                        | Underfitting                                                                                                                         | Model Capacity                                                                                         |
|------|------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| 6    | <b>Model Complexity</b>      | Very complex models (e.g., deep neural networks with many layers or parameters).                                   | Very simple models (e.g., linear regression used for non-linear data).                                                               | Depends on architecture: linear models (low), neural networks (high), decision trees (medium to high). |
| 7    | <b>Visual Representation</b> | Model creates a <b>wavy curve</b> or overly complex surface that passes through every data point, including noise. | Model draws a <b>simple straight line</b> or smooth surface that misses real trends in data.                                         | Capacity allows the model to bend and twist its decision boundary.                                     |
| 8    | <b>Causes</b>                | - Too many parameters - Small dataset - Training too long - No/low regularization                                  | - Too few parameters - Insufficient training - High regularization - Oversimplified model                                            | Determined by the architecture of the model and training process.                                      |
| 9    | <b>Real-world Example</b>    | A facial recognition model trained only on studio-lit photos performs poorly in real-world, low-light conditions.  | A weather prediction model using only temperature as input may not capture complex dependencies like humidity, wind, or seasonality. | A good weather model must have enough capacity to include multiple factors without overfitting.        |
| 10   | <b>Solutions</b>             | - Apply regularization (L1/L2) - Use dropout - Use cross-validation - Reduce model size - Increase dataset size    | - Increase model complexity - Reduce regularization - Add more relevant features - Train for more epochs                             | Choose model with <b>suitable capacity</b> based on dataset size, complexity, and performance goals.   |